International Conference on Data-Processing and Networking (ICDPN-2026)

Date: 25th -26th September, 2026

ORGANISED BY : Institute of Technology and Business in České Budějovice, Near Prague, Czech Republic, Europe (Venue).

********** CALL FOR PAPERS **********

SPECIAL SESSION ON

Agentic AI for Healthcare Systems: Autonomous Systems for Intelligent Care Delivery

SESSION ORGANIZERS:

Shaurya Gupta, Associate Professor, UPES Dehradun India, Email: shaurya.gupta@ddn.upes.ac.in
Sonali Vyas, Professor, JK Lakshmipat University, Jaipur, India Email: sonali.vyas@jklu.edu.in
Sourabh Singh Verma, Associate Professor, IoT & IS Department, Manipal University Jaipur, Jaipur. Email: sourabhsingh.verma@jaipur.manipal.edu

SESSION DESCRIPTION:

The rapid advancement of digital technologies, particularly web-based platforms and intelligent systems, has significantly transformed the healthcare ecosystem. In recent years, the emergence of Agentic AI—AI systems capable of autonomous decision-making, goal-driven behaviour, and multi-agent collaboration—has opened new frontiers in delivering smart, adaptive, and efficient healthcare solutions. This special session on "Agentic AI in Healthcare: Autonomous Systems for Intelligent Care Delivery" aims to explore cutting-edge research and innovations in the design, development, and deployment of agent-based AI systems in healthcare. Agentic AI enables the integration of diverse technologies into cohesive, intelligent frameworks capable of real-time decision-making, personalised treatment planning, and proactive healthcare management.

The session will provide a platform to review global trends, challenges, and advancements while identifying future research directions for building scalable, trustworthy, and ethical agent-based healthcare systems. It aims to benefit researchers, academicians, healthcare professionals, industry practitioners, and students engaged in interdisciplinary healthcare innovation.

RECOMMENDED TOPICS:

Topics to be discussed in this special session include (but are not limited to) the following:

- Multi-Agent Systems in Healthcare
- Autonomous Clinical Decision Support Systems
- Intelligent Patient Monitoring using AI Agents
- Bio-Inspired and Adaptive Agent Architectures
- Agent-based IoT Healthcare Applications
- Distributed Healthcare Systems and Edge AI
- Healthcare Workflow Automation using Agentic AI

Edit with WPS Office

- Security, Ethics, and Trust in Autonomous Healthcare Systems
- Big Data Integration in Agent-driven Healthcare
- Disaster and Emergency Response using AI Agents
- Cloud-based Agent Frameworks for Healthcare
- Interoperability and System Integration in Smart Healthcare

SUBMISSION PROCEDURE:

Researchers and practitioners are invited to submit papers for this special theme session on **Agentic AI for Healthcare Systems: Autonomous Systems for Intelligent Care Delivery** ***on or before*** **3oth JULY 2026.** All submissions must be original and may not be under review by another publication. INTERESTED AUTHORS SHOULD CONSULT THE CONFERENCE'S GUIDELINES FOR MANUSCRIPT SUBMISSIONS at **https://www.icdpn-conf.com/Downloads**. All submitted papers will be reviewed on a double-blind, peer review basis.

NOTE: While submitting paper in this special session, please specify **Agentic AI for Healthcare Systems: Autonomous Systems for Intelligent Care Delivery** at the top (above the paper title) of the first page of your paper.

****

Edit with WPS Office